

Albert (Geyang) Xu

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RESEARCH INTERESTS

- Data-centric robustness of ML pipelines; ML/MLOps reliability & monitoring; Responsible & transparent data management; Streaming & distributed data systems; Data provenance & explanations; Governance of ML systems

EDUCATION

University of California, San Diego Sep 2022 - Mar 2024
M.S. in Computer Science (GPA: 3.88/4.0) La Jolla, CA

- Courses: Operating Systems, Networks, Programming Languages, ML, Probabilistic Modeling, Data Ethics

University of Liverpool Oct 2020 - Jun 2022
B.S. in Computer Science (GPA: 3.93/4.0, WES Approved, top 3%) Liverpool, UK

Xi'an Jiaotong-Liverpool University Sep 2018 - Jun 2020
B.S. in Information and Computing Science (Years 0–1 under the UoL 2+2 Programme) Suzhou, China

PUBLICATIONS

- **PVLDB'25: Stress-Testing ML Pipelines with Adversarial Data Corruption (SAVAGE)**. *PVLDB 18(11): 4668–4681 (2025)*. Preprint: <https://arxiv.org/abs/2506.01230> · Code: <https://github.com/lodino/savage>

RESEARCH EXPERIENCE

UCSD Halicioğlu Data Science Institute Oct 2023 - May 2025
Research Engineer La Jolla, CA

- Built a modular “Injector” pipeline (pattern generation → sampling → injection → evaluation) with **beam-search** pruning and **Optuna/TPE-based Bayesian tuning** to systematically surface structured corruptions that reduce downstream AUC by **>0.25** compared to random-parameter attacks
- Co-authored **SAVAGE**, designing **corruption dependency graphs** and **bi-level black-box** search to model mechanism-aware missingness, selection-bias, and outlier patterns for **pipeline-level stress-testing** of ML systems
- Ran **Inject → Clean → Retrain** benchmarks across missing-value, selection-bias, and outlier scenarios to evaluate state-of-the-art cleaning, debiasing, and UQ pipelines and expose their **data-centric robustness gaps**

INDUSTRY EXPERIENCE

Fuyao Glass America Inc. Oct 2025 - Present
Manufacturing Data Analyst Dayton, OH

- Consolidated production data from machines and inspection systems into a clean analytics layer using **Python** and **SQL ETL**, then delivered role-based dashboards that track OEE, FPY, scrap, and changeover in near real time
- Ran set-point experiments on magnetron sputtering and tempering using **A/B** and **DOE**, governed changes with **Git** schema-migrations, SOPs, and parameter logs, and delivered reproducible **Jupyter** notebooks
- Hardened reliability via access control, code review, cron and alerts, **99.3%** job success, **2.4 h/week** less downtime

4Pexonic Inc. Oct 2024 - Oct 2025
Back-end Engineer San Diego, CA

- Developed an academic impact **analysis platform** tracking citation patterns across **tens of thousands of records** using Node.js, MongoDB, and Next.js with **Docker**, serving university researchers and policy institutions
- Built two-tier caching with in-process **LRU** and **Redis** for bibliometric queries, reducing median and p95 latency by **40%** for complex research trend analysis and cross-disciplinary impact calculations
- Integrated **gRPC** microservices for data processing pipelines and implemented secure authentication with **bcrypt**/session management, configured **Nginx SSL** to protect academic publication data and researcher information

TECHNICAL SKILLS

Programming Languages: Python, Java, Go, SQL, C/C++, JavaScript, R

Frameworks and Tools: Docker, Git, gRPC, Node.js, Next.js, Django, React, MongoDB, Redis, Jupyter, Nginx, AWS